
The Selfless-tron

On electrons, networks, and the mathematics that separates
prosocial geometry from selfish deliberation

An essay for mathematics and physics teachers

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The electron does not deliberate and nor does it survey the circuit, weigh its options, and choose the path of least resistance. And yet the circuit settles, without exception and without instruction, into precisely the configuration that a mathematician would describe as the minimum of a certain sum over its edges, the same sum whose minimum is what Kirchhoff's laws, applied locally at every junction, happen to select. That local and global should coincide so completely reveals something very deep about the structures of mathematics that are capable of describing these physical laws, and we teach neither as well as we might.

1. The Curriculum We Have and the One We Could Have

There is a version of secondary school mathematics and physics that a student could complete entirely without ever understanding what a network is, why parallel structures behave differently from serial ones, or why the number most suited to averaging a set of rates is not the one obtained by adding them up and dividing by how many there are. This is, broadly speaking, the version that exists.

The arithmetic mean is introduced early and stays late, clean and computable, and wrong for perhaps a third of the situations in which students are asked to apply it. The harmonic mean, which is what you need when averaging speeds over equal distances, combining resistors in parallel, finding the effective stiffness of springs in series, or characterising the hydraulic conductance of a pipe network, appears if at all as a footnote in statistics, stripped of the physical intuition that would make it comprehensible and memorable.

Network theory, in the sense of understanding how flow is distributed across connected structures subject to conservation laws, is almost entirely absent from the pre-university curriculum, despite being the mathematical language of electrical engineering, fluid dynamics, traffic planning, epidemiology, supply chains, and the internet.

The case here is that the curriculum is incomplete in a particular prescient way in our highly networked system world, and that the connections it fails to draw are among the most illuminating available to a student of mathematics or physics.

The curriculum teaches students to calculate averages, combine resistors, solve spring problems and sketch reciprocal graphs, but it rarely tells them that these are all versions of the same question: what quantity actually adds? Once that question is asked, the harmonic mean, conductance, compliance, network flow and even the Braess paradox become parts of one curriculum story rather than disconnected tricks.

2. The Arithmetic Mean Is Not the Average

When teachers say “find the average,” they almost always mean the arithmetic mean, and the habit is so entrenched that it survives into professional life, producing the celebrated error in which a car travelling equal distances at 60 and 40 kilometres per hour is assigned an average speed of 50 when the correct answer is 48. The error is conceptual, and it belongs to a category of error that the present curriculum is poorly equipped to prevent, because it never clearly explains what the **arithmetic mean** is actually computing or why

it is sometimes the wrong computation to perform.

The arithmetic mean is the right tool when the quantities being averaged contribute equally in an additive sense: the typical height in a class, the mean temperature over a week, the average mass of a sample. For rates, ratios, or quantities whose reciprocals add, the additive structure assumed by the arithmetic mean is not the structure of the problem. The average speed of the car travelling equal distances is the single constant speed that would produce the same total journey time, which works out to $2v_1v_2 / (v_1 + v_2)$, the harmonic mean of the two speeds.

Students who learn only the arithmetic mean are being given one member of a family of averages without being told that the family exists or that the choice of member is determined by the structure of the problem. The geometric mean is the correct average for quantities that compound multiplicatively. The quadratic mean governs quantities that accumulate as squares, which is why it appears in the definition of AC voltage and in the kinetic theory of gases. The harmonic mean governs rates and reciprocal additions.

A curriculum that introduced all four means, and connected each to the physical and economic situations in which it naturally arises, would give students something the present one withholds, which is the sense that mathematical definitions are responses to the structure of real problems rather than arbitrary conventions to be memorised.

The harmonic mean is a little less straightforward to calculate than the arithmetic mean being the arithmetic mean applied to reciprocals, then inverted. That is three elementary steps, and what makes it feel exotic is not its complexity but its unfamiliarity, which is purely a curriculum choice.

3. The Natural Quantity and the Named Quantity

There is a pattern running through every case in which the harmonic mean appears, and it is worth making explicit before moving to the physical applications, because once it is seen it cannot be unseen.

In every instance, the quantity we name and teach is the reciprocal of the quantity that actually composes additively in the relevant physical or mathematical structure. We name *speed* and teach students to divide it when averaging over distances, when the quantity that adds naturally is *slowness*: the time taken to traverse a unit of distance. Average slowness over a journey, then invert to recover average speed. We name *resistance* and wire it into Ohm's law as $V = IR$, when the quantity that adds in a parallel circuit is *conductance* $G = 1/R$, and the form that makes the physics legible is $I = GV$: current is conductance

times voltage, flow is ease-of-flow times driving difference. We name *stiffness* k and define it as $k = F/x$, when the quantity that adds in a series spring system is *compliance* $1/k$: the extension per unit force, the ease with which the spring yields.

The naming choices are not arbitrary and they are not wrong. Resistance is pedagogically useful because it scales with difficulty, which matches intuition, and the historical formulation $V = IR$ is too entrenched to be replaced. But the naming choices carry a cost: they make the formula for combining elements in the problematic direction, parallel resistors or series springs, look like a peculiar exception requiring its own rule, when it is in fact a straightforward arithmetic operation on the natural quantity, followed by an inversion. Once a student sees that $G_{\text{eff}} = G_1 + G_2$ for parallel resistors and $1/k_{\text{eff}} = 1/k_1 + 1/k_2$ for series springs are the *same statement* in different notation, the harmonic mean formula ceases to be an awkward special case and becomes the expected consequence of averaging something that adds additively and then converting back to the named quantity.

The return-journey speed problem is the most transparent instance of this structure, and it is worth tracing it piece by piece. The natural quantity is time per kilometre, the pace of the journey, and it adds arithmetically over equal-distance segments: the total time is $d/v_1 + d/v_2$, the sum of the paces multiplied by the shared distance. The average pace is $(1/v_1 + 1/v_2)/2$, a straightforward arithmetic mean of the reciprocals. The average speed is then the reciprocal of the average pace:

$$\bar{v} = \frac{1}{\frac{1/v_1 + 1/v_2}{2}} = \frac{2}{\frac{1}{v_1} + \frac{1}{v_2}} = \frac{2v_1v_2}{v_1 + v_2}.$$

The three steps are: flip each speed to its pace, take the arithmetic mean of the paces, flip back to a speed. This is the harmonic mean, and it is not a special formula memorised for a particular type of problem; it is the mechanical consequence of working with the right quantity, which happens to be the reciprocal of the named one.

Domain	Named quantity	Natural quantity	Combining rule
Average speed	speed v	pace $1/v$ (time per km)	$\bar{v} = \frac{2v_1v_2}{v_1 + v_2}$
Parallel resistors	resistance $R = V/I$	conductance $G = 1/R$ ($I = GV$)	$G_{\text{eff}} = G_1 + G_2$
Series springs	stiffness $k = F/x$	compliance $1/k$ ($x = F/k$)	$\frac{1}{k_{\text{eff}}} = \frac{1}{k_1} + \frac{1}{k_2}$

Figure 2. The curriculum often names the quantity on the left, but the quantity that composes arithmetically is often its reciprocal. Once the natural quantity is identified, the apparently special formula becomes ordinary addition followed, where needed, by conversion back to the named quantity.

Figure 2. In each domain, the curriculum names the quantity on the left but the quantity that composes arithmetically, the one whose mean is easy to compute, is on the right. The harmonic mean of the named quantity is the arithmetic mean of the natural quantity, inverted. Teaching $I = GV$ alongside $V = IR$ is not an additional complication; it is the clarification that makes the parallel-combination formula self-evident rather than a rule to be memorised.

Pre-calculus, this pattern is already available as a lesson in rates and their reciprocals. A rate measures how much of something you get per unit of something else: kilometres per hour, amperes per volt, newtons per metre. Its reciprocal measures how much of the something-else you need per unit of the something: hours per kilometre, ohms, metres per newton. When a process is governed by equal allocations of the something-else rather than equal allocations of the something itself, the natural average is computed in the reciprocal domain and then converted back. This observation requires no calculus, no differential equations, and no formal definition of a derivative; it requires only a clear understanding of what a rate is and what it means to take its reciprocal, both of which are within reach of any student who has learned to convert between speed and journey time.

4. The Hidden Unity: Springs, Circuits, and Pipes

One of the genuinely elegant facts of applied mathematics is that springs in series, resistors in parallel, and pipes in parallel are all governed by exactly the same equation, and that equation is the harmonic mean. A student who learns Hooke's law in physics, Ohm's law in a circuits module, and the Hagen-Poiseuille equation in fluid dynamics, assuming they are fortunate enough to encounter all three, will almost certainly do so in separate lessons, from separate teachers, in separate years, with no indication that the formula being manipulated is structurally identical in each case.

The underlying reason for this unity is conservation. In a series spring system, the same force is transmitted through every spring; the extensions add, so the compliances add, giving an effective stiffness with the structure of a harmonic mean. In a parallel resistor circuit, the same voltage appears across every resistor; the currents add, so the conductances add, and the effective resistance follows the same formula. In a parallel pipe network, the same pressure difference drives flow through every pipe; the volumetric flow rates add, the hydraulic conductances add, and the effective fluid resistance is again a harmonic mean expression.

In each case, a conservation law forces the reciprocal quantity to be additive in the parallel configuration and the direct quantity to be additive in the serial configuration. This is not a coincidence but the same mathematical structure appearing across three physical domains

because the domains share the same topological relationship between their elements, and a teacher who pauses to say so is doing something that no textbook does adequately: demonstrating that physical laws are not a collection of unrelated formulas but recurring themes in a single underlying composition.

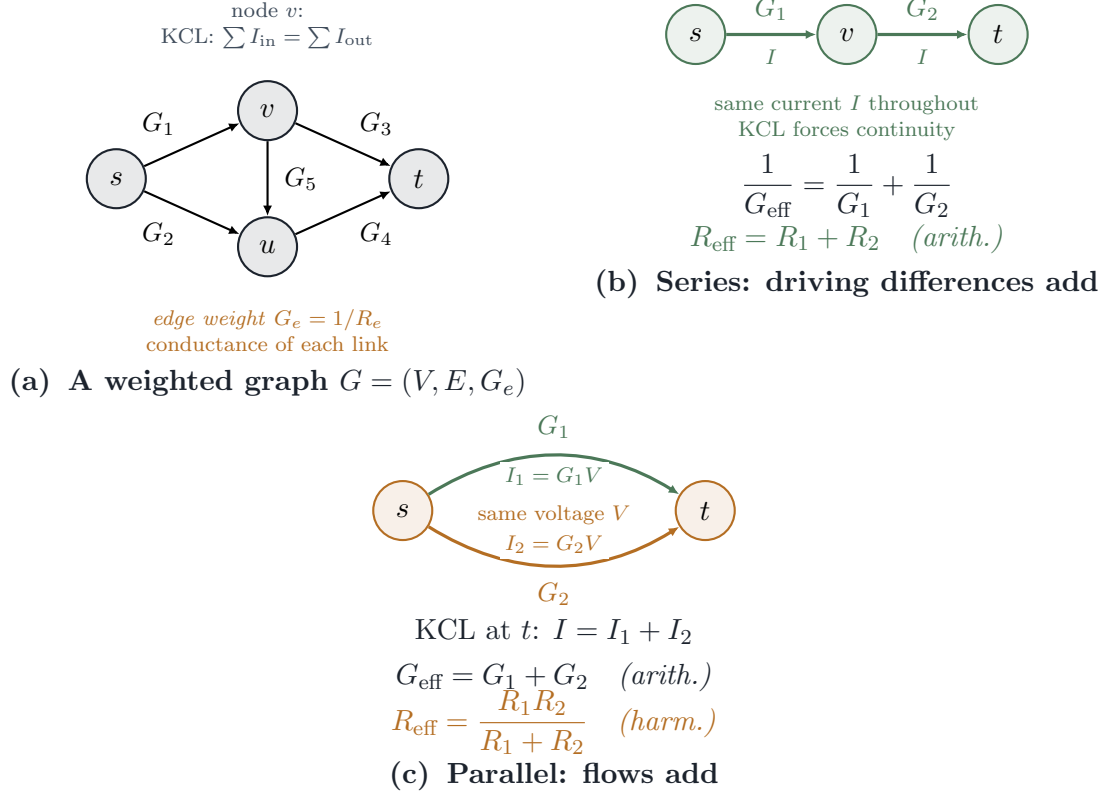


Figure 1. A network is a graph $G = (V, E)$ in which each edge e carries a conductance weight $G_e = 1/R_e$. KCL at every internal node v constrains currents to balance. In a series chain (b), the same current passes each edge and driving differences accumulate arithmetically. In a parallel arrangement (c), the same driving difference V is shared and currents accumulate arithmetically in conductance, yielding the harmonic-mean expression for the combined resistance. The substitution $\{V, I, R\} \rightarrow \{\Delta P, Q, R_f\}$ or $\{F, x, 1/k\}$ produces the identical structure for pipes and springs respectively.

5. Conservation Laws and the Appearance of Purpose

There is a peculiarity in the behaviour of passive networks worth dwelling on, because it points toward something philosophically important that the curriculum tends to pass over in silence.

When current flows through a resistor network, it distributes itself across the available paths in exactly the proportions that satisfy Kirchhoff's laws at every node, and Thomson's

theorem tells us that this distribution is also, as a mathematical consequence of those linear constraints, the unique one that minimises the total power $\sum_e I_e^2 R_e$ subject to the same boundary conditions. The minimisation is not a physical cause but a mathematical equivalence: Kirchhoff flow and minimum-dissipation flow are two names for the same object, one described locally, the other globally, and the electron does not choose between them any more than a triangle chooses to have angles summing to 180° . The theorem is a fact about convex geometry, and it holds for linear resistors only; a transistor with current-dependent resistance breaks the linearity on which it rests, and a superconductor makes it vacuous by setting the dissipation to zero regardless of how the current is distributed.

What makes the pattern remarkable is not the circuit but the company it keeps. Light travelling from one medium into another refracts at precisely the angle that minimises total travel time between source and destination, Fermat's principle, first articulated in the seventeenth century and still remarkable in its implications. A mechanical system evolving under classical dynamics follows the trajectory that makes the integral of the Lagrangian stationary, Hamilton's principle of stationary action, which in quantum mechanics becomes the statement that amplitudes from different paths add constructively only where the action is at a saddle, selecting the classical trajectory by interference rather than by intention. Soap bubbles adopt the surface of minimal area consistent with their boundary conditions, solving a problem in the calculus of variations without possessing anything we would recognise as intention.

These systems behave, in the language of the philosophical literature, teleologically: as if directed toward a goal, as if physical processes had purposes. The deeper truth, which Lagrangian and Hamiltonian mechanics eventually made precise, is that the variational formulation is not a separate principle layered on top of the differential equations but an equivalent reformulation, and it reveals something the differential formulation conceals: that physical systems can be understood, yes as evolving forward in time according to local rules, but also as selecting, from all possible configurations, the one that satisfies a global criterion, and that in the linear passive case these two descriptions are not merely equivalent but identical.

Maupertuis called it the principle of least action and believed it was evidence of divine economy in the design of the universe. We now know it is a mathematical theorem rather than a theological claim, but the sense of wonder he felt at the apparent purposefulness of physical law is epistemically appropriate, and we do students no favours by not exposing them to it before they have a chance to feel it.

6. From Selfless Atoms to Selfish Agents: The Braess Paradox

The most striking illustration of what is lost when networks are absent from the curriculum is the Braess paradox, because it requires network thinking to state, game theory to explain, and produces a conclusion that directly contradicts the physical intuition students develop from passive networks.

In a passive network, adding a parallel path always helps. Conductances add; resistance falls; flow increases. This is a theorem, unconditional and exceptionless. In an active network populated by agents who make individual decisions, adding a new path can make every agent worse off. The paradox, described by Dietrich Braess in 1968 and observed subsequently in road networks in Stuttgart, Seoul, and New York, demonstrates that a new high-capacity road can increase equilibrium travel time for all users, because the individually rational response channels everyone onto a route that saturates the most congested links simultaneously.

The reason is that traffic is not electrons. Electrons distribute across all available paths at once, weighted by conductance, without competition, settling into the globally optimal configuration as a consequence of Kirchhoff's laws. Drivers each take the path that minimises their own journey time, without internalising the congestion cost they impose on others, and the resulting Wardrop equilibrium, the traffic analogue of the Nash equilibrium, guarantees only that no individual can improve by switching routes unilaterally, a local stability condition that says nothing about whether the collective outcome is good.

This contrast, passive particles obeying global variational principles against active agents trapped in locally stable but globally suboptimal equilibria, is one of the central intellectual distinctions in applied mathematics, the distinction between physics and economics, and a curriculum that teaches it gives students a vocabulary for understanding phenomena that range from server congestion and vaccination policy to the prisoner's dilemma and the tragedy of the commons.

7. What Would Change

The case being made here is not for a radical restructuring of the curriculum but for a set of connections that are already latent in the material and need only to be made explicit.

When students encounter the formula for resistors in parallel, the teacher can note that this is the harmonic mean of the resistances, halved, that the same formula governs springs

in series and pipes in parallel, and that the reason it recurs is conservation. When students are introduced to averages, all four means can be presented with a concrete example distinguishing them, the lesson being that the choice of mean is determined by the structure of the problem. When students encounter Snell's law, the question worth asking is not what the formula says but why light should behave this way at all, and the answer, that this local rule at the boundary is equivalent to a global variational statement about the entire trajectory, opens a door to ideas that the curriculum otherwise reaches only in university-level physics. When students study traffic in geography or economics, the difference between Kirchhoff flow and Wardrop flow is worth a paragraph, and the name for what happens when adding a road makes everyone slower is worth knowing.

None of these additions require new textbooks or extended syllabi. They require teachers to be given time and permission to draw the connections that the curriculum currently obscures by separating them into distinct subjects, distinct years, and distinct examination papers.

The world is made of networks, and students who learn to see them, who understand that springs and circuits and pipes and roads are instances of the same underlying mathematics, that conservation laws produce what looks like purposeful behaviour, and that the difference between selfless atoms and selfish agents runs all the way from Kirchhoff to Keynes, are better equipped to think about almost everything that matters.

The same reciprocal thinking appears in Boyle's law, inverse-square gravitational or Coulomb potentials. The point being that students should learn to ask what quantity is naturally additive before choosing a formula. See appendix for more reciprocal riffs.

8. A Note on Difficulty and Motivation

A predictable objection is that these ideas are too difficult for secondary school students. This objection underestimates students and overestimates the difficulty of the ideas. The harmonic mean involves reciprocals, which are already on every secondary mathematics syllabus. The observation that a formula appears in multiple physical contexts because the underlying conservation structure is shared does not require calculus to appreciate. The variational principle that light takes the fastest path is a motivating story that can precede the formal statement of Snell's law rather than following it as a postscript. The distinction between selfless electrons and selfish drivers is conceptually accessible to any student old enough to have sat in a traffic jam.

What these ideas require is intellectual curiosity. Students who are told that the formula on the board appears, in the same form, across circuits, pipes, and springs, because conservation laws have a particular mathematical structure, are being offered a reason to care about the formula that extends beyond the examination. The evidence from educational psychology is fairly consistent: intrinsic motivation, of the kind produced by genuine intellectual engagement with ideas, produces more durable learning than instrumental motivation of the kind produced by examination pressure, and that observation applies as much to the manifesto as to the mathematics.

9. Conclusion: The Selfless-tron

The essay began with an anthropomorphism, and it is worth being precise about what it can honestly claim. The electron does not know the harmonic mean, and Thomson's theorem is not a statement about efficiency or thermodynamic purpose; it is the observation that for a linear passive resistor network, the distribution satisfying Kirchhoff's laws at every node is the same object, described globally, as the distribution minimising $\sum_e I_e^2 R_e$ subject to the same boundary conditions. The electron produces the minimum-dissipation configuration not because it is seeking anything but because Kirchhoff flow and minimum-dissipation flow are two descriptions of the same geometry, one local, one variational, forced to coincide by the convexity of the constraint space. A purely resistive network wastes every joule of electrical input as heat; the theorem says only that the distribution of that inevitable waste is the smallest it could be, not that the network is in any useful sense efficient.

The anthropomorphism is nonetheless worth keeping, because what it actually says is more interesting than a naive reading suggests. Local rules, applied consistently at every junction without any global instruction, produce the globally structured distribution that the harmonic mean describes, and the wonder is precisely that local and global coincide so completely in the linear passive case and diverge so dramatically when the network is populated by agents with preferences. The contrast with Wardrop flow is therefore sharper than the earlier sections suggested: Kirchhoff achieves global optimality through local conservation because the physics is passive and linear; Wardrop fails to achieve it because the agents are adaptive and self-interested, and the mathematics of the two cases, convex optimisation on one side, game theory on the other, are as different as the outcomes they produce.

What the electron knows, then, is that local enforcement of conservation at every node is sufficient, with no further instruction, to produce the globally structured distribution that the harmonic mean describes. The same structure appears in fluid flow, in mechanical springs, in the trajectory of light through glass, in the stationary-action principle, and in

the potential field between Earth and Moon. These correspondences are evidence that certain mathematical structures, variational principles, conservation laws, the algebra of reciprocal addition, are genuinely fundamental, and a curriculum that makes them visible, even at the level of pattern recognition rather than formal proof, gives students something more durable than any individual formula.

There is a further reflection that the essay's argument invites, one that reaches beyond the curriculum into something rather older. Richard Dawkins argued in *The Selfish Gene* that apparent altruism at the level of the organism is explicable by, and in some sense reducible to, selfishness at the level of the gene: the organism sacrifices because its genes benefit, and the unit of selection is not the individual but the replicator. It is a powerful inversion, and it has shaped fifty years of thinking about the relationship between individual behaviour and collective outcomes.

The physics described in this essay runs in precisely the opposite direction. The elementary particle is structurally prosocial, not as a consequence of selection pressure or kin relatedness or repeated interaction, but geometrically, necessarily, because the mathematics of conservation forces it to distribute itself across all available paths simultaneously, weighted by conductance, without preference and without competition. The electron is the altruist by constitution. It is only at the level of aggregated, organised, preferring matter, the organism, the driver, the investor routing capital, that selfishness emerges, and when it does it produces the Braess paradox: a degradation of collective outcomes that serves no gene, no replicator, no deeper organisational logic, but simply the locally rational calculation of a mind that can model its own interest and not the interest of the network.

The Dawkins story is one of selfishness all the way down, with altruism as an emergent illusion at higher levels. The network story told here is the inverse: prosociality at the bottom, with selfishness as an emergent property of the level at which preferences and deliberation appear. Whether this is consoling or alarming is a question the mathematics cannot answer. What it can say is that the universe, at the level of electrons and photons and soap films, does not defect. Defection is an achievement of complexity, and it costs everyone something.

Dawkins named his central actor for what it does at the level of the replicator: preserving itself through the organism that carries it. *TRON*¹ stars Jeff Bridges as a programmer drawn inside a computer system, thus imagines the opposite: a human drawn inside the machine of circuitry pathways. Indeed we are no longer [only] biological machines carrying our selfish gene being often memed agents inside artificial computational networks. The protagonist of this essay is naturally then a tron, but not the pro-social pro-tron, not a

¹*TRON*, directed by Steven Lisberger, Walt Disney Productions, 1982. The film stars Jeff Bridges as Kevin Flynn, a programmer who is drawn inside a computer system, where programs, circuits and games become the world itself. See Disney, "TRON," <https://movies.disney.com/tron>.

positively charged emblem of goodwill but rather the elec-tron the carrier whose negative charge make it self-less not selfish, whose local obedience produces the global order of the circuit.

The world is made of networks, and students who learn to see them, who understand that springs and circuits and pipes and roads are instances of the same underlying mathematics, that local conservation produces what looks like global purpose, and that the difference between selfless atoms and selfish agents runs all the way. from Kirchhoff to Keynes, are better equipped to think about almost everything that matters.

The point is that students need better questions. What is conserved? What adds? What is the natural reciprocal? Is the network passive or strategic? Does local behaviour produce a global optimum, or merely a stable failure? A curriculum that teaches those questions would give students a grammar for circuits, springs, pipes, traffic, epidemics, and markets.

10. Appendix: Further Mathematical Riffs

There is a further observation that the curriculum leaves unmade, and it concerns the shape of the curves students are asked to plot but rarely asked to interpret.

When a physics class plots extension against load for a spring, the convention places the independent variable, the applied force, on the horizontal axis and the dependent response, the extension, on the vertical. The gradient is $\Delta x / \Delta F$, which is the compliance $1/k$: the natural quantity, the one that adds in a series system, emerging directly from the standard graph without any algebraic inversion. The curriculum then defines the spring constant as the reciprocal of this gradient, names it stiffness, and proceeds to use $F = kx$, inverting the relationship that the graph makes most visible. The choice is defensible but instructive: we teach the slope of the graph, which is compliance, then immediately discard it in favour of its reciprocal, which is stiffness, and present the combination rule for series springs as a formula requiring explanation, when it is simply the statement that compliances add.

Gas laws present a similar situation. The relationship $PV = nRT$ is taught as a product law, but the relationship between pressure and volume at constant temperature, $V \propto 1/P$, is a reciprocal law, and its graph is a hyperbola: one of those asymptotically approaching curves that the mathematics syllabus tends to treat as exercises in curve-sketching rather than as instances of a physical pattern. The harmonic mean of two pressures P_1 and P_2 is the pressure at which a gas would have a volume equal to the arithmetic mean of the volumes at those two pressures, a fact that takes thirty seconds to derive from $PV = \text{const}$

and that connects an algebraic definition to a geometric property of the hyperbola, but this connection is not made.

Electric and gravitational potentials carry the same $1/r$ structure. The potential due to a point mass or charge falls off as the reciprocal of distance, and between two such sources the total potential is a sum of two $1/r$ terms. This is worth pausing on, because it affords a genuine pre-calculus riff on what the harmonic mean might mean in space.

Consider the Earth and the Moon separated by distance d , with a test mass placed at a point along the line between them at distance r from the Earth, so that it is at distance $d - r$ from the Moon. The gravitational potential at that point, taking the convention that potential is negative and increases toward zero at large distances, is:

$$V(r) = -\frac{GM_E}{r} - \frac{GM_M}{d-r}.$$

This is a sum of two $1/r$ terms, and it is not monotone: it has a local maximum somewhere between the two bodies, the saddle point at which a test mass placed exactly there would, in the absence of the system's rotation, feel zero net gravitational force. That saddle point, the first **Lagrange point** L1, is where the attractive pull of the Earth toward smaller r is exactly balanced by the pull of the Moon toward larger r .

The condition for L1 in the simplified non-rotating case is $GM_E/r^2 = GM_M/(d-r)^2$, which gives $r/\sqrt{M_E} = (d-r)/\sqrt{M_M}$, a relationship determined by the geometric mean of the masses rather than the harmonic mean. But the potential at L1 itself, and more generally the equipotential surfaces between the two bodies, have a structure that rewards interpretation in terms of the natural quantities. The quantity $1/r$ is the natural variable for a $1/r$ potential: it is the *gravitational pace*, the potential per unit mass per unit gravitational constant, and the total potential is a weighted arithmetic sum in these natural variables. An equipotential surface between Earth and Moon is defined by:

$$\frac{GM_E}{r} + \frac{GM_M}{d-r} = \text{const},$$

which, in the case of equal masses $M_E = M_M = M$, reduces to $1/r + 1/(d-r) = \text{const}$, and the point where this sum is minimised, the saddle of the potential, occurs at $r = d/2$, the arithmetic mean of zero and d , by symmetry. The harmonic mean enters when one asks what effective single distance from a combined source would produce the same total potential as two equal masses, each at their respective positions: the answer is $H(r, d-r)/2 = r(d-r)/d$, the harmonic mean of the two distances halved, a compact expression for what it would cost in potential energy to be located between the two bodies rather than at one of them.

This is merley a riff: an invitation to notice that the same mathematical object that determines the effective spring constant of a series system and the combined resistance

of a parallel circuit turns up when you ask how deep a potential well is dug between two gravitational sources. The $1/r$ potential is a reciprocal law, and the natural quantities for reciprocal laws are the reciprocals themselves, and the average of a reciprocal law over a spatial interval is a harmonic mean. The curriculum already contains $V \propto 1/r$ and $F = kx$ and $V = IR$ and $PV = \text{const}$; what it lacks is the observation that these are all instances of the same reciprocal structure, and that the harmonic mean is the right tool for combining any of them.

A student who has seen this pattern once, concretely, in the spring-stiffness or the return-journey problem, and who is then shown the gravitational potential between Earth and Moon and asked what the natural variable is, is being offered an experience that is rare in secondary education: the discovery that a thing they understood in one domain applies, unchanged in structure, to a **domain** they had not previously connected to it. That experience is what mathematical education is supposed to procure.